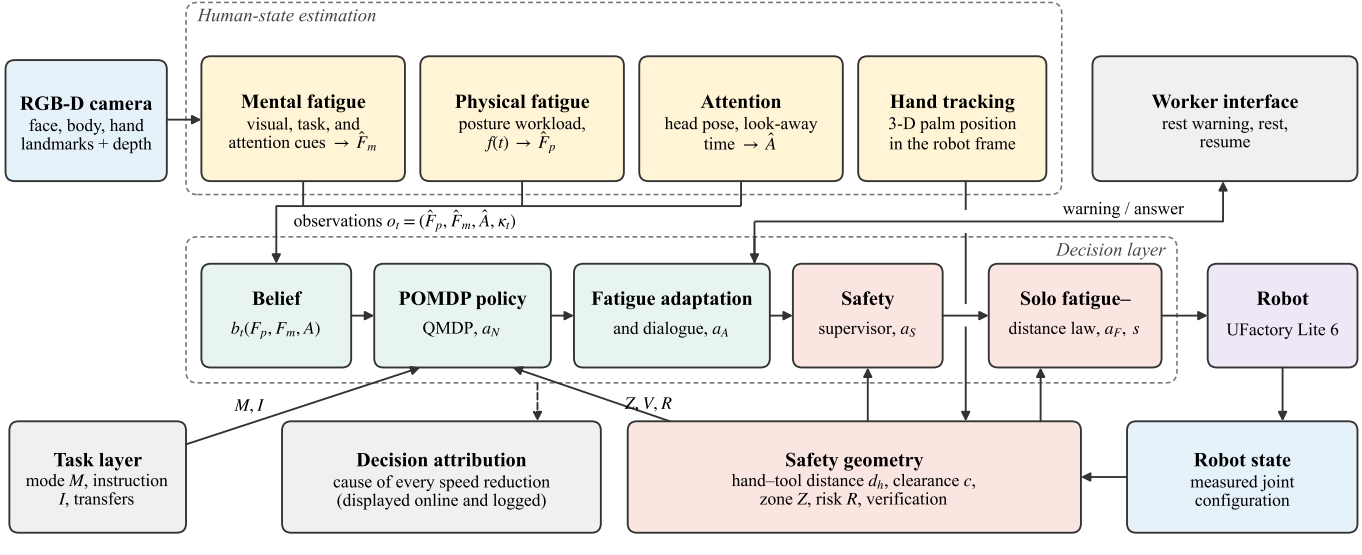

Human-state estimation
RGB-D camera
face, body, hand landmarks + depth
Mental fatigue
visual, task, and attention cues → $\hat{F}_m$
Physical fatigue
posture workload, $f(t) \rightarrow \hat{F}_p$
Attention
head pose, look-away time → $\hat{A}$
Hand tracking
3-D palm position in the robot frame
Worker interface
rest warning, rest, resume
warning / answer
observations $o_t = (\hat{F}_p, \hat{F}_m, \hat{A}, \kappa_t)$
Decision layer
Belief
$b_t(F_p, F_m, A)$
POMDP policy
QMDP, a_N
Fatigue adaptation
and dialogue, a_A
Safety
supervisor, a_S
Solo fatigue–
distance law, a_F, s
Robot
UFactory Lite 6
M, I
Z, V, R
Task layer
mode M, instruction I, transfers
Decision attribution
cause of every speed reduction (displayed online and logged)
Safety geometry
hand–tool distance d_h, clearance c, zone Z, risk R, verification
Robot state
measured joint configuration